

STATISTICS FOR DATA SCIENCE

Formula Sheet + Exam-Friendly Wording

Prepared from all 10 unique July Intake class-slide PDFs (505 slides reviewed).

How to use this sheet

Each entry has the concept meaning, standard mathematical notation, a keyboard-safe exam version using ordinary characters, a sentence explaining how to read it, a real-world use, and the exact class-slide reference. Always define your variables and finish calculations with a conclusion in context.

Lecturer-aligned wording evidence

The recurring assessment instruction says to answer in your own words and use formulas or calculations where appropriate (slide 13 in the session decks). Lecturer-supplied text equivalents are retained, including conditional probability as $P(\text{intersection})$ divided by $P(\text{independent event})$ (Unit 1, Session 2, slide 20), population variance as the sum of $(x \text{ minus mean})^2$ divided by N (Unit 2, Session 5, slide 48), and Chi-square with observed and expected frequency written out (Unit 3, Session 8, slide 49). Wording is expanded only where needed to remain clear and typeable.

Reference format: Unit - Session - slide number.

UNIT 1 - PROBABILITY

Basic probability

Meaning: Measures how likely an event is to occur.

Standard notation: $P(A) = n(A) / n(S)$

Keyboard-safe exam version: $P(A) = n(A) / n(S)$

Read it as: Probability of A equals the number of outcomes in event A divided by the total number of outcomes in the sample space.

Real-world example: A bag has 10 sweets and 3 are red. Use this formula to find the chance of picking a red sweet.

Symbols: $n(A)$ = favourable outcomes; $n(S)$ = all possible outcomes.

Unit 1 - Session 1 - slide 23

Expected number

Meaning: Estimates how many times an event should occur over repeated trials.

Standard notation: **Expected number = $n \times P(A)$**

Keyboard-safe exam version: `expected_number = n * P(A)`

Read it as: Expected number equals the number of future trials multiplied by the probability of the event.

Real-world example: If 2 out of every 10 customers buy juice, use this to guess how many of the next 100 customers may buy juice.

Symbols: n = number of trials.

Unit 1 - Session 1 - slide 28

Probability range and total

Meaning: States the validity rules every probability distribution must satisfy.

Standard notation: **$0 \leq P(A) \leq 1$; $\sum P(\text{outcomes}) = 1$**

Keyboard-safe exam version: `0 <= P(A) <= 1; sum_of_all_probabilities = 1`

Read it as: A probability must be between zero and one, inclusive; the probabilities of all possible outcomes add up to one.

Real-world example: Before using a probability table, check that no chance is below 0 or above 1 and that all the chances add to 1.

Unit 1 - Session 1 - slide 33; Session 3 - slides 22 and 26

Complement rule

Meaning: Finds the chance that an event does not occur.

Standard notation: **$P(A') = 1 - P(A)$**

Keyboard-safe exam version: $P(\text{not_}A) = 1 - P(A)$

Read it as: Probability that A does not happen equals one minus the probability that A happens.

Real-world example: If the chance of rain is 0.3, use this to find the chance that it will not rain.

Unit 1 - Session 1 - slides 32-33

Independent-event multiplication

Meaning: Finds the chance that two unrelated events both occur.

Standard notation: **$P(A \cap B) = P(A) \times P(B)$**

Keyboard-safe exam version: $P(A_and_B) = P(A) * P(B)$

Read it as: For independent events, probability of A and B equals probability of A multiplied by probability of B.

Real-world example: Use this for two events that do not affect each other, such as tossing a coin and rolling a die.

Unit 1 - Session 1 - slide 31

Dependent-event multiplication

Meaning: Finds the chance that two related events both occur.

Standard notation: $P(A \cap B) = P(A) \times P(B|A)$

Keyboard-safe exam version: $P(A_and_B) = P(A) * P(B_given_A)$

Read it as: For dependent events, probability of A and B equals probability of A multiplied by probability of B given A.

Real-world example: Use this when the first choice changes the next one, such as taking two sweets from a bag without putting the first one back.

Unit 1 - Session 1 - slide 31

Addition rule

Meaning: Finds the chance that at least one of two events occurs.

Standard notation: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Keyboard-safe exam version: $P(A_or_B) = P(A) + P(B) - P(A_and_B)$

Read it as: Probability of A or B equals probability of A plus probability of B minus probability of both A and B.

Real-world example: Use this to find the chance of choosing a red sweet or a green sweet.

Exam cue: If mutually exclusive, the intersection is zero, so just add $P(A)$ and $P(B)$.

Unit 1 - Session 1 - slide 32

Conditional probability

Meaning: Finds the chance of an event after another event is already known.

Standard notation: $P(A|B) = P(A \cap B) / P(B)$

Keyboard-safe exam version: $P(A_given_B) = P(A_and_B) / P(B)$

Read it as: Probability of A given B equals probability of A and B divided by probability of B.

Real-world example: Use this to find the chance that a learner plays soccer when you already know the learner plays a sport.

Symbols: B is the condition already known to have occurred.

Unit 1 - Session 2 - slides 19-20

Cumulative distribution function

Meaning: Gives the probability accumulated up to a specified value.

Standard notation: $F(x) = P(X \leq x)$

Keyboard-safe exam version: $F(x) = P(X \leq x)$

Read it as: The cumulative probability at x equals the probability that random variable X is less than or equal to x.

Real-world example: Use this to find the chance that a child waits 5 minutes or less for the bus.

Unit 1 - Session 3 - slide 25

Expected value of a discrete variable

Meaning: Gives the long-run average outcome of a random variable.

Standard notation: $E(X) = \sum[x \times P(X=x)]$

Keyboard-safe exam version: $E(X) = \sum(x * P(X=x))$

Read it as: Expected value equals the sum of each possible value multiplied by its probability.

Real-world example: Use this to find the average number of goals a team may score over many games.

Unit 1 - Session 3 - slides 27-28; recap in Unit 2 - Session 5 - slide 15

UNIT 2 - DESCRIPTIVE STATISTICS

Arithmetic mean

Meaning: Gives the numerical average of a dataset.

Standard notation: $\bar{x} = \Sigma x / n$

Keyboard-safe exam version: $\bar{x} = \text{sum}(x) / n$

Read it as: The mean equals the sum of all observations divided by the number of observations.

Real-world example: Add five test marks and divide by five to find the learner's average mark.

Symbols: $\bar{x}$ = sample mean; n = number of observations.

Unit 2 - Session 4 - slides 24-26; Session 5 - slide 48

Mean from a frequency table

Meaning: Calculates an average when values are presented with frequencies.

Standard notation: $\bar{x} = \Sigma(fx) / \Sigma f$

Keyboard-safe exam version: $\bar{x} = \text{sum}(f \cdot x) / \text{sum}(f)$

Read it as: The mean equals the sum of frequency times value divided by the total frequency.

Real-world example: Use this when a table shows how many learners received each test mark.

Symbols: f = frequency; x = data value.

Unit 2 - Session 4 - slide 39

Median position

Meaning: Locates the middle observation after sorting the data.

Standard notation: **Odd n : $(n+1)/2$; Even n : average of positions $n/2$ and $(n/2)+1$**

Keyboard-safe exam version: odd n : $(n+1)/2$; even n : average of positions $n/2$ and $(n/2)+1$

Read it as: Arrange the data from smallest to largest. For an odd count, take the middle value. For an even count, average the two middle values.

Real-world example: Use this to find the middle house price or middle pocket-money amount when a few values are very large.

Unit 2 - Session 4 - slides 28-29

Empirical mean-median-mode relationship

Meaning: Estimates one central measure from the other two for a moderately skewed distribution.

Standard notation: **Mode = $3 \times \text{Median} - 2 \times \text{Mean}$**

Keyboard-safe exam version: $\text{mode} = 3 \cdot \text{median} - 2 \cdot \text{mean}$

Read it as: Mode equals three times the median minus two times the mean.

Real-world example: Use this to estimate the most common value when the mean and median are already known.

Exam cue: Use only when the question supplies or implies this empirical relationship.

Unit 2 - Session 4 - slides 32-33

Range

Meaning: Measures total spread from the smallest to largest value.

Standard notation: **Range = maximum - minimum**

Keyboard-safe exam version: $\text{range} = \text{maximum} - \text{minimum}$

Read it as: Range equals the largest value minus the smallest value.

Real-world example: If the lowest mark is 40 and the highest is 90, the range shows the full gap between them.

Unit 2 - Session 5 - slide 37

Interquartile range

Meaning: Measures the spread of the middle fifty percent of the data.

Standard notation: $IQR = Q3 - Q1$

Keyboard-safe exam version: $IQR = Q3 - Q1$

Read it as: Interquartile range equals the third quartile minus the first quartile.

Real-world example: Use this to see how spread out the middle half of a class's marks are.

Unit 2 - Session 5 - slides 37 and 43

Quartile deviation

Meaning: Measures half of the interquartile spread around the median.

Standard notation: $QD = (Q3 - Q1) / 2$

Keyboard-safe exam version: $QD = (Q3 - Q1) / 2$

Read it as: Quartile deviation equals the third quartile minus the first quartile, divided by two.

Real-world example: Use this to describe how far the middle half of the marks spreads around the middle.

Unit 2 - Session 5 - slide 37

Outlier fences

Meaning: Identifies unusually low or high observations using quartiles.

Standard notation: **Lower** = $Q1 - 1.5 \times IQR$; **Upper** = $Q3 + 1.5 \times IQR$

Keyboard-safe exam version: lower = $Q1 - 1.5 \times IQR$; upper = $Q3 + 1.5 \times IQR$

Read it as: The lower fence equals Q1 minus one point five times the IQR; the upper fence equals Q3 plus one point five times the IQR. Values outside the fences are outliers.

Real-world example: Use this to spot a value that looks unusually large or small, such as one learner taking much longer than everyone else.

Unit 2 - Session 5 - slides 30 and 43

Population variance

Meaning: Measures average squared distance from the population mean.

Standard notation: $\sigma^2 = \sum(x - \mu)^2 / N$

Keyboard-safe exam version: population_variance = $\text{sum}((x - \text{mean})^2) / N$

Read it as: Population variance equals the sum of each value minus the population mean, squared, divided by the population size.

Real-world example: Use this when you have every learner's mark in the whole class and want to measure how different the marks are.

Symbols: μ = population mean; N = population size.

Unit 2 - Session 5 - slides 38 and 48

Sample variance

Meaning: Estimates variability from sample data using sample size minus one.

Standard notation: $s^2 = \sum(x - \bar{x})^2 / (n - 1)$

Keyboard-safe exam version: sample_variance = $\text{sum}((x - \bar{x})^2) / (n - 1)$

Read it as: Sample variance equals the sum of each value minus the sample mean, squared, divided by sample size minus one.

Real-world example: Use this when you only have marks from a few learners and want to estimate how spread out all learners' marks are.

Symbols: s^2 = sample variance.

Unit 2 - Session 5 - slides 38 and 45

Standard deviation

Meaning: Measures typical spread around the mean in the original data units.

Standard notation: $\sigma = \sqrt{\text{variance}}$ or $s = \sqrt{\text{variance}}$

Keyboard-safe exam version: `standard_deviation = sqrt(variance)`

Read it as: Standard deviation equals the square root of the variance.

Real-world example: Two learners may have the same average score. The one with the smaller standard deviation has more consistent scores.

Exam cue: Smaller standard deviation means greater consistency.

Unit 2 - Session 5 - slides 38, 44 and 48

Coefficient of variation

Meaning: Compares relative variability as a percentage of the mean.

Standard notation: $CV = (\text{standard deviation} / \text{mean}) \times 100\%$

Keyboard-safe exam version: `CV = (standard_deviation / mean) * 100`

Read it as: Coefficient of variation equals standard deviation divided by the mean, multiplied by one hundred percent.

Real-world example: Use this to fairly compare the spread of two things measured on different scales, such as height and weight.

Exam cue: Smaller CV means more consistency; use CV to compare relative variability across different scales.

Unit 2 - Session 5 - slides 38, 49 and 61

Covariance

Meaning: Shows whether two variables tend to move together or in opposite directions.

Standard notation: $\text{Cov}(X,Y) = \Sigma[(x-x_{\text{bar}})(y-y_{\text{bar}})] / \text{divisor}$

Keyboard-safe exam version: population: `cov = sum((x-x_bar)*(y-y_bar))/N`; sample: divide by `(n-1)`

Read it as: Covariance equals the sum of paired deviations from their means divided by the appropriate population or sample divisor.

Real-world example: Use this to see whether ice-cream sales and temperature usually rise together.

Exam cue: Use N for a population and n minus one for a sample.

Unit 2 - Session 5 - slides 54 and 57

Pearson correlation

Meaning: Measures the direction and strength of a linear relationship.

Standard notation: $r = \text{Cov}(X,Y) / (s_x \times s_y)$

Keyboard-safe exam version: `r = covariance / (SD_x * SD_y)`

Read it as: Pearson correlation equals the covariance of X and Y divided by the product of their standard deviations.

Real-world example: Use this to measure how strongly hot weather and ice-cream sales move together. It does not prove that one causes the other.

Exam cue: r ranges from minus one to plus one; correlation does not prove causation.

Unit 2 - Session 5 - slides 54 and 56-59

UNIT 3 - INFERENCE STATISTICS

Standard error of the sample mean

Meaning: Measures how much sample means are expected to vary between samples.

Standard notation: $SE = \sigma / \sqrt{n}$

Keyboard-safe exam version: $SE = \text{population_SD} / \text{sqrt}(n)$

Read it as: Standard error equals the population standard deviation divided by the square root of the sample size.

Real-world example: Use this to judge how close the average from a small group may be to the true average for everyone.

Unit 3 - Session 6 - slides 23, 25 and 29-30

One-sample z statistic

Meaning: Standardises the difference between a sample mean and claimed mean when population spread is known.

Standard notation: $z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$

Keyboard-safe exam version: $z = (\text{x_bar} - \mu_0) / (\text{population_SD} / \text{sqrt}(n))$

Read it as: Z equals the sample mean minus the hypothesised population mean, divided by the population standard deviation over the square root of sample size.

Real-world example: A cereal factory says boxes weigh 500 grams and knows the usual spread. Use a z-test to check the claim.

Symbols: Use when population standard deviation is known; commonly with a large sample.

Unit 3 - Session 7 - slides 27 and 42; Session 9 - slide 26

One-sample t statistic

Meaning: Tests one sample mean against a claimed mean when population spread is unknown.

Standard notation: $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$

Keyboard-safe exam version: $t = (\text{x_bar} - \mu_0) / (\text{sample_SD} / \text{sqrt}(n))$

Read it as: T equals the sample mean minus the hypothesised population mean, divided by the sample standard deviation over the square root of sample size.

Real-world example: A teacher says the class average is 70, but only a few marks are available. Use a t-test to check the claim.

Symbols: Use when population standard deviation is unknown; $df = n - 1$.

Unit 3 - Session 7 - slides 24-25 and 30-32

Independent two-sample t statistic

Meaning: Tests whether two separate groups have different population means.

Standard notation: $t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1) + (s_2^2/n_2)}$

Keyboard-safe exam version: $t = (\text{x1_bar} - \text{x2_bar}) / \text{sqrt}((s1^2/n1) + (s2^2/n2))$

Read it as: T equals the difference between the two sample means divided by the square root of sample one variance over its size plus sample two variance over its size.

Real-world example: Use this to compare the average marks of Class A and Class B when they contain different learners.

Symbols: Deck formula uses the unpooled standard-error form; the deck lists $df = n_1 + n_2 - 2$ for its equal-variance version.

Unit 3 - Session 7 - slides 25-26 and 34-35

Paired t statistic

Meaning: Tests the mean change for matched or before-and-after observations.

Standard notation: $t = \bar{d} / (s_d / \sqrt{n})$

Keyboard-safe exam version: `t = d_bar / (SD_of_differences / sqrt(n))`

Read it as: T equals the mean of the paired differences divided by the standard deviation of the differences over the square root of the number of pairs.

Real-world example: Use this to compare the same learners' marks before and after extra lessons.

Symbols: $\bar{d}$ = mean paired difference; $df = n - 1$.

Unit 3 - Session 7 - slides 25-26 and 36-37

P-value decision

Meaning: Determines whether the sample evidence is strong enough to reject the null hypothesis.

Standard notation: If $p < \alpha$: reject H_0 ; if $p \geq \alpha$: fail to reject H_0

Keyboard-safe exam version: `if p_value < alpha: reject H0; otherwise: fail_to_reject H0`

Read it as: If the p-value is less than the significance level, reject the null hypothesis. Otherwise, fail to reject the null hypothesis.

Real-world example: Use the p-value like a decision signal: a very small p-value tells you the result is unlikely to be just luck.

Exam cue: Write 'fail to reject', not 'accept', unless your lecturer explicitly requires different wording.

Unit 3 - Session 7 - slide 17; Session 8 - slides 18-19; Session 9 - slides 24, 26 and 28

Confidence and significance

Meaning: Connects the confidence level with the accepted Type I error risk.

Standard notation: Confidence level = $1 - \alpha$

Keyboard-safe exam version: `confidence_level = 1 - alpha`

Read it as: Confidence level equals one minus the significance level.

Real-world example: A 95 percent confidence level means accepting a 5 percent risk of making a Type I error.

Unit 3 - Session 6 - slide 41

Chi-square test statistic

Meaning: Measures how far observed categorical counts differ from expected counts.

Standard notation: $\chi^2 = \sum [(O - E)^2 / E]$

Keyboard-safe exam version: `chi_square = sum((O-E)^2 / E)`

Read it as: Chi-square equals the sum, over all categories, of observed frequency minus expected frequency squared, divided by expected frequency.

Real-world example: Use this to check whether learners really choose four ice-cream flavours equally often.

Symbols: O = observed frequency; E = expected frequency.

Unit 3 - Session 8 - slides 25, 28 and 49

Expected frequency - contingency table

Meaning: Calculates the count expected in a table cell if the variables are independent.

Standard notation: $E = (\text{row total} \times \text{column total}) / \text{grand total}$

Keyboard-safe exam version: `E = (row_total * column_total) / grand_total`

Read it as: Expected frequency equals row total multiplied by column total, divided by the grand total.

Real-world example: Use this to work out how many learners you would expect in each box of a two-way table if the two choices were unrelated.

Unit 3 - Session 8 - slides 25-26 and 49

Expected frequency - equal proportions

Meaning: Calculates the count expected in each category when categories are equally likely.

Standard notation: $E = \text{total observations} / \text{number of categories}$

Keyboard-safe exam version: $E = \text{total_observations} / \text{number_of_categories}$

Read it as: Expected frequency equals total observations divided by the number of categories.

Real-world example: If 100 sweets should be split equally among four colours, the expected number is 25 of each colour.

Unit 3 - Session 8 - slides 25-26, 33, 35 and 49

Chi-square degrees of freedom

Meaning: Selects the appropriate Chi-square reference distribution.

Standard notation: **Goodness of fit:** $df = k-1$; **Independence/homogeneity:** $df = (r-1)(c-1)$

Keyboard-safe exam version: GOF: $df = k-1$; table: $df = (r-1)*(c-1)$

Read it as: For goodness of fit, degrees of freedom equals number of categories minus one. For independence or homogeneity, multiply rows minus one by columns minus one.

Real-world example: Use this number to choose the correct row in the Chi-square table.

Unit 3 - Session 8 - slides 25, 32 and 49

Simple regression line

Meaning: Predicts the dependent variable from an independent variable.

Standard notation: $\hat{Y} = a + bX$

Keyboard-safe exam version: $Y_hat = a + b*X$

Read it as: Predicted Y equals the intercept plus the slope multiplied by X.

Real-world example: Use hours studied to predict a learner's test mark.

Symbols: a = intercept; b = slope.

Exam cue: The deck example is Y equals twenty plus four X.

Unit 3 - Session 9 - slides 18 and 20

UNIT 4 - LINEAR ALGEBRA

Vector addition

Meaning: Combines vectors by adding matching components.

Standard notation: $\mathbf{v} + \mathbf{w} = (v_1 + w_1, v_2 + w_2, \dots)$

Keyboard-safe exam version: $v_plus_w = (v1+w1, v2+w2, \dots)$

Read it as: Add vectors component by component.

Real-world example: If a game character moves 3 steps right and then 2 more, vector addition combines the movements.

Unit 4 - Session 10 - slide 21

Scalar multiplication

Meaning: Changes a vector's size by multiplying every component by one number.

Standard notation: $k\mathbf{v} = (kv_1, kv_2, \dots)$

Keyboard-safe exam version: $k_times_v = (k*v1, k*v2, \dots)$

Read it as: Multiply every component of the vector by the scalar k.

Real-world example: If every movement in a game must be doubled, multiply every vector part by 2.

Unit 4 - Session 10 - slide 21

Dot product

Meaning: Combines matching vector components into one scalar value.

Standard notation: $\mathbf{v} \cdot \mathbf{w} = \sum (v_i w_i)$

Keyboard-safe exam version: $\text{dot}(v, w) = v1*w1 + v2*w2 + \dots = \text{sum}(v_i*w_i)$

Read it as: Dot product equals the sum of matching vector components multiplied together.

Real-world example: Use this to score how closely a person's skills match the skills needed for a job.

Unit 4 - Session 10 - slides 21 and 27

Vector magnitude

Meaning: Calculates the length or size of a vector.

Standard notation: $|\mathbf{v}| = \sqrt{\sum (v_i^2)}$

Keyboard-safe exam version: $\text{magnitude_v} = \text{sqrt}(v1^2 + v2^2 + \dots)$

Read it as: Magnitude equals the square root of the sum of the squared vector components.

Real-world example: Use this to find the total size of a movement or the overall strength of a group of scores.

Unit 4 - Session 10 - slides 25-27

Angle between vectors

Meaning: Measures the direction difference between two vectors.

Standard notation: $\cos \theta = (\mathbf{A} \cdot \mathbf{B}) / (|\mathbf{A}||\mathbf{B}|)$

Keyboard-safe exam version: $\cos(\text{theta}) = \text{dot}(A, B) / (\text{magnitude_A} * \text{magnitude_B})$

Read it as: Cosine of the angle equals the dot product divided by the product of the two vector magnitudes.

Real-world example: Use this to see whether two movement arrows point in similar or different directions.

Unit 4 - Session 10 - slide 27

Cosine similarity

Meaning: Measures how similar two vectors are by their direction.

Standard notation: $\text{cosine similarity} = (\mathbf{A} \cdot \mathbf{B}) / (|\mathbf{A}||\mathbf{B}|)$

Keyboard-safe exam version: $\text{cosine_similarity} = \text{dot}(A, B) / (\text{magnitude_A} * \text{magnitude_B})$

Read it as: Cosine similarity equals the dot product of A and B divided by the product of their magnitudes.

Real-world example: A music app can use this to find two listeners with similar song choices.

Exam cue: One means same direction, zero means orthogonal, and minus one means opposite directions.

Unit 4 - Session 10 - slides 29-31

Matrix multiplication

Meaning: Combines matrices through row-by-column products.

Standard notation: $C_{ij} = \sum(a_{ik} b_{kj})$

Keyboard-safe exam version: $C_{ij} = \text{sum}(a_{ik} * b_{kj})$ for $k=1$ to n

Read it as: Each entry in the product equals the sum of row elements from A multiplied by matching column elements from B.

Real-world example: A game can use this to change many points at once when an object is moved, stretched, or rotated.

Unit 4 - Session 10 - slide 39

Transpose

Meaning: Rearranges a matrix by changing rows into columns.

Standard notation: A^T : rows become columns

Keyboard-safe exam version: $A_{\text{transpose}} = \text{rows_of_A_changed_to_columns}$

Read it as: The transpose of A is obtained by interchanging its rows and columns.

Real-world example: Use this to swap rows and columns, like turning a marks table so subjects become rows instead of columns.

Unit 4 - Session 10 - slides 40 and 44

Two-by-two determinant

Meaning: Produces a scalar that indicates whether a matrix is invertible.

Standard notation: For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det(A) = ad - bc$

Keyboard-safe exam version: $\det_A = a*d - b*c$

Read it as: For a two-by-two matrix, determinant equals a times d minus b times c.

Real-world example: Use this to check whether a matrix can be reversed. If the determinant is 0, it cannot be reversed.

Exam cue: An inverse exists only when the determinant is not zero.

Unit 4 - Session 10 - slides 41 and 44

Two-by-two inverse

Meaning: Produces a matrix that reverses the effect of the original matrix.

Standard notation: $A^{-1} = \frac{1}{(ad-bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Keyboard-safe exam version: $A_{\text{inverse}} = (1/(a*d-b*c)) * \begin{bmatrix} d, -b \\ -c, a \end{bmatrix}$

Read it as: The inverse equals one divided by the determinant, multiplied by the matrix formed by swapping a and d and changing the signs of b and c.

Real-world example: Use this to undo a matrix change or solve two equations with two unknown numbers.

Unit 4 - Session 10 - slides 42 and 44

Trace

Meaning: Adds the entries on the main diagonal of a square matrix.

Standard notation: $\text{tr}(A) = \text{sum of main-diagonal entries}$

Keyboard-safe exam version: $\text{trace_A} = \text{sum}(\text{main_diagonal_entries})$

Read it as: The trace equals the sum of the entries on the main diagonal.

Real-world example: Add the numbers on the main diagonal of a square table to get one quick summary number.

Unit 4 - Session 10 - slides 43-45

EXAM DECISION AND WORDING GUIDE

Choosing a test

Z-test: use for a population mean when the population standard deviation is known (the slides also associate it with large samples). One-sample t-test: compare one sample mean with a claimed value when population standard deviation is unknown. Independent t-test: compare means from two separate groups. Paired t-test: compare the same people or items before and after. Chi-square: use categorical counts for goodness of fit, independence, or homogeneity. ANOVA: compare three or more means.

Unit 3 - Session 7 - slides 20-27 and 40-41; Session 8 - slides 23-25; Session 9 - slides 18-20

Safe conclusion templates

Reject H_0 : At the stated significance level, there is sufficient evidence to conclude that [state the alternative in the context of the question].

Fail to reject H_0 : At the stated significance level, there is insufficient evidence to conclude that [state the alternative in context].

Association: There is / is not a statistically significant association between [variable 1] and [variable 2].

Goodness of fit: The observed frequencies do / do not differ significantly from the expected frequencies.

Unit 3 - Session 7 - slides 17, 32, 35, 37 and 43; Session 8 - slides 29, 33 and 35

Common notation translated

Keyboard form	Meaning
sum(...)	the sum of
sqrt(n)	the square root of n
x_bar	the sample mean
mu	the population mean
s	the sample standard deviation
sigma	the population standard deviation
H_0 / H_1	the null hypothesis / the alternative hypothesis
alpha	the significance level
df	degrees of freedom
AND / OR	intersection / union
given	given that

Final check: identify the formula, substitute values, show the result, then interpret the result in the language of the scenario.